

Architecture of 8086 Microprocessor Family

Q. Describe the internal architecture of the 8086 Microprocessor. Explain the functions of the Bus Interface Unit (BIU), Execution Unit (EU), Registers, and Memory Segmentation.

[DEF] Definition to Remember

The Intel 8086 is a 16-bit microprocessor with a 16-bit data bus and a 20-bit address bus (can address 1 MB of memory). Its internal architecture is divided into two parallel functional units: the Bus Interface Unit (BIU) for memory/bus operations, and the Execution Unit (EU) for instruction execution, enabling early pipelining.

1. Bus Interface Unit (BIU)

Handles all data and address transfers on the system buses. The EU relies entirely on the BIU to fetch data.

2. Execution Unit (EU)

Responsible for decoding and executing instructions.

3. Registers of 8086 (16-bit)

Category	Registers
General Purpose	`AX` (Accumulator), `BX` (Base), `CX`
Pointer & Index	`SP` (Stack Pointer), `BP` (Base Point
Segment Registers	`CS` (Code Segment), `DS` (Data Segmen
Special	`IP` (Instruction Pointer), **Flag Reg

4. Memory Segmentation

The 8086 has 16-bit registers but needs to access 20-bit addresses (1 MB). Memory is logically divided into segments of 64 KB each.

[MUST-WRITE] Must-Write Points (for 10 marks)

- 8086 is a 16-bit processor; 16-bit data bus; 20-bit address bus (1 MB memory).
- Divided into two units: Bus Interface Unit (BIU) and Execution Unit (EU).
- BIU: Handles bus operations, calculates addresses, manages a 6-byte Instruction Queue (Pipelining).
- EU: Decodes and executes instructions fetched from the BIU queue using the ALU.
- Registers: General Purpose (AX, BX, CX, DX), Segment (CS, DS, SS, ES), Pointers (SP, BP, IP), Flags.
- Memory Segmentation: Divides 1 MB memory into 64 KB logical segments.
- Physical Address Calculation: $(\text{Segment} * 10H) + \text{Offset}$ (solves 20-bit address using 16-bit registers).

[RECALL] Quick Recall

8086 -> 16-bit Data, 20-bit Address (1MB) -> BIU (Bus/Memory, 6-byte Queue, Address Calc) + EU (Decode/Execute, ALU) -> Registers (AX, BX, Segments, Pointers) -> Segmentation (Physical Addr = Segment*10H + Offset)